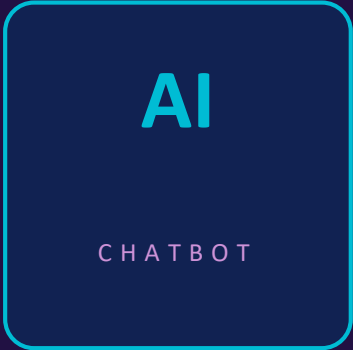


A dark blue rounded square with a teal border. Inside, the letters 'AI' are in large teal font, and 'CHATBOT' is in smaller white font below it.

AI

CHATBOT

Chatbot System

in Machine Learning

Working, Algorithms, Models & Backend Workflow of AI Chatbots

Introduction to Chatbot System

Hello! How can I help you?

I need order status info.

Sure! Your order is on its way.



AI Chatbot

What is a Chatbot System?

A chatbot system is an AI-based software that interacts with users through text or voice and is designed to simulate human-like conversation using Artificial Intelligence technologies such as Machine Learning and Natural Language Processing (NLP).

Machine Learning

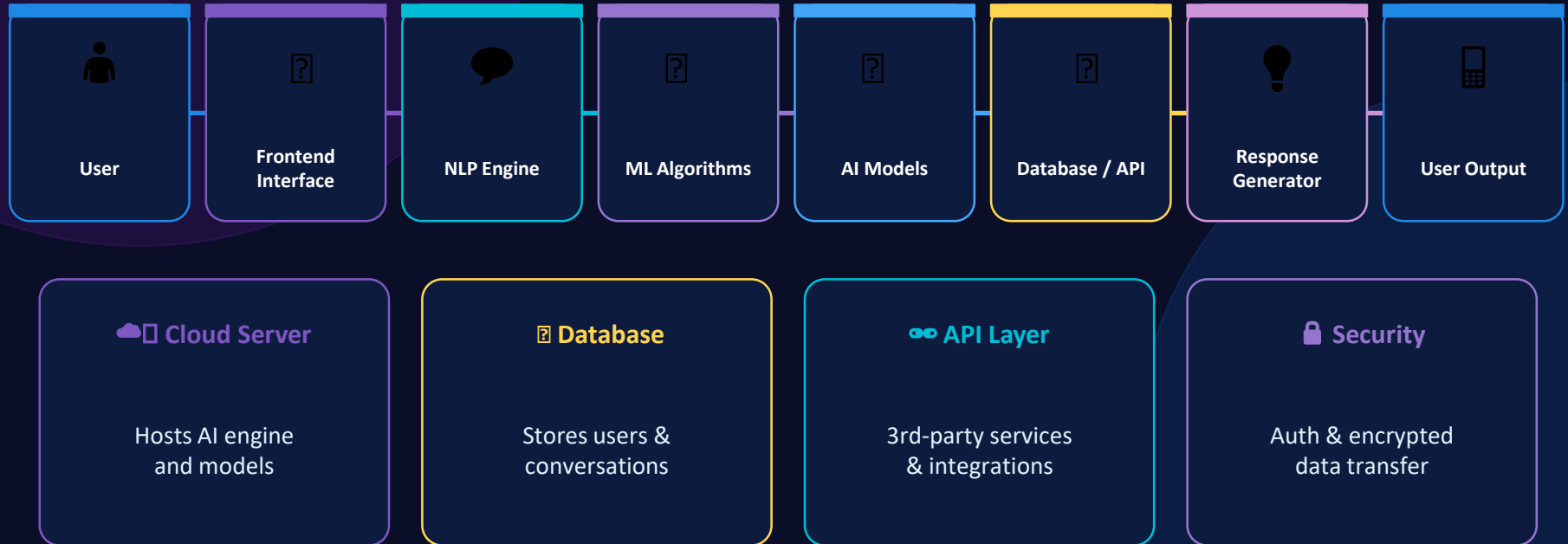
NLP

AI

Text & Voice

Human-like

Workflow of Chatbot System



Types of Chatbots



Rule-Based Chatbot

Follows pre-defined rules and decision trees. Responds only to specific keywords or commands. Limited but reliable for structured tasks.



AI-Based Chatbot

Uses Machine Learning and NLP to understand natural language. Learns from conversations and improves over time. Powers advanced chatbots like GPT.



Voice Assistant Chatbot

Interacts using speech recognition and text-to-speech. Examples: Siri, Alexa, Google Assistant. Converts voice to text and processes with AI.



Hybrid Chatbot

Combines rule-based logic with AI capabilities. Falls back to rules when AI is uncertain. Best of both approaches for enterprise use.



Rule-Based Chatbot

Definition

Responds to users according to predefined rules, patterns, and decision-making logic. Can only answer questions that match its programmed rules.

How It Works

User enters message → Chatbot searches keywords → Compares with rules → Selects matching rule → Displays predefined response.

Decision Tree

Works like a flowchart. Users guided through options. Common in customer support. Example: Account Issue → Reset Password.

Pattern Matching

Matches user input with stored patterns. Returns predefined answer. Example: Pattern: Hello → Response: Hi! How can I help?

IF-THEN Logic

Simple programming rules. IF user says Hi THEN reply Hello! IF message contains refund THEN ask for order number.

Features

✓ Easy to build ✓ Fast response ✓ Low cost ✓ Good for FAQs ✗ Cannot learn ✗ Limited conversation



AI-Based Chatbot

Definition

Understands human language, learns from interactions, and provides intelligent responses using Artificial Intelligence and Machine Learning.

How It Works

User enters message → NLP analyzes meaning → ML model processes → Chatbot predicts best response → System improves over time.

NLP Processing

Helps chatbot understand human language. Identifies meaning, intent, and key words. Example: Book flight to Lahore tomorrow → Action=Book, Destination=Lahore.

ML Algorithms

Logistic Regression for intent classification. Decision Trees for prediction. Naive Bayes for text classification. Random Forest for accuracy.

Deep Learning

Uses artificial neural networks. RNNs understand word sequences. Transformers understand context. Used in ChatGPT and Gemini.

Example

User: I am stressed about exams. AI Chatbot: Exam stress is common. Would you like study tips or stress-management advice?



Voice Assistant Chatbot

Definition

Intelligent system that uses speech recognition, NLP, and AI to communicate with users through voice commands.

Step 1 - Voice Input

User speaks a command. Example: What is the weather today? The system captures the audio signal from the microphone.

Step 2 - Speech Recognition

System converts spoken words into text. Identifies words, accents, and language patterns accurately.

Step 3 - NLP Processing

Chatbot understands the meaning and intent of the command. Extracts key information from the converted text.

Step 4 - AI Processing

Machine learning models determine the best response. Context and history of conversation are considered.

Step 5 - Text to Speech

Response is converted back into spoken words using TTS technology. Examples: Siri, Alexa, Google Assistant.



Hybrid Chatbot

Definition

Combines rule-based logic and artificial intelligence to provide both accurate predefined responses and intelligent conversational abilities.

How It Works

User sends message → Rule engine checks for match → If rule found: predefined response. If no rule: AI model processes → Intelligent response.

Rule-Based Components

Decision Trees for structured questions. IF-THEN Rules for specific commands. Keyword Matching for common queries and FAQs.

AI Components

Machine Learning for understanding complex queries. NLP for natural language understanding. Neural Networks and Deep Learning models.

Advantages

Best of both worlds. Reliable for known questions via rules. Intelligent for unknown queries via AI. Ideal for enterprise applications.

Use Cases

Customer service chatbots. Banking assistants. Healthcare appointment systems. E-commerce product support bots.

Technologies Used in Chatbot Systems



Machine Learning

Trains chatbots on data to predict and classify user intents



Natural Language Processing

Parses and understands human language text and speech



Deep Learning

Multi-layer neural nets for advanced language understanding



Neural Networks

Brain-inspired models that recognize patterns in data



APIs

Connect chatbot to external services, data, and platforms



Cloud Computing

Scalable infrastructure for hosting and deploying AI models



Databases

Store conversations, user data, and knowledge base content

Algorithms Used in Chatbot Systems

Naive Bayes

→ Text classification & intent prediction

Support Vector Machine (SVM)

→ Classifies user messages & intents

Decision Tree

→ Decision making & response prediction

Neural Network Algorithm

→ Learns patterns from user data

Random Forest

→ Improves accuracy using multiple trees

Transformer Algorithm

→ Understands language context in advanced AI

K-Nearest Neighbor

→ Finds similar user questions

Machine Learning Models Using In Chatbot System

GPT

Generative Pre-trained
Transformer

- ▶ Human-like response generation

BERT

Bidirectional Encoder
Representations

- ▶ Language understanding & QA

RNN

Recurrent Neural
Network

- ▶ Conversation memory & sequence learning

LSTM

Long Short-Term
Memory

- ▶ Long-range context in conversations

ANN

Artificial Neural
Network

- ▶ Pattern recognition

CNN

Convolutional Neural
Network

- ▶ Image-based chatbot systems

Machine Process in Chatbot System



Conclusion

AI chatbots use Machine Learning, NLP, algorithms, and advanced models to understand human language and provide intelligent responses.

Machine Learning

Trains the core prediction engine

NLP Processing

Enables language understanding

Smart Algorithms

Powers intent & response logic

Advanced Models

GPT, BERT, LSTM for deep context